

1. Irreversible reactions

- ① $A \rightarrow B \rightarrow R_{AB} = -k_b$
 ② $A \rightarrow C \rightarrow R_{AC} = -k_c$ } homog, 0th
 ③ $A \rightarrow F \rightarrow R_{AF} = -k_F C_A \rightarrow$ homog, 1st

Stationary film: $v=0$ and $n_A'' = j_A$
 thickness: L

$y=L$: impermeable (no diffusion across)
 unreactive (no reaction occurring at this boundary)

$y=0$: conc $A = C_0$

Determine C_A

Stationary media:

$$n_A'' = j_A = -p D_{AB} \nabla n_A$$

$$N_A'' = J_A^* = -C D_{AB} \nabla x_A$$

$$D_{AB} \frac{d^2 C_A}{dx^2} + \dot{N}_A = 0$$

zero order: $\dot{N}_A = k_0 \rightarrow$ Reaction 1+2

First order: $\dot{N}_A = k_1 C_A \rightarrow$ Reaction 3

$$D_{AB} \frac{d^2 C_A}{dy^2} - k_F C_A = 0 \quad m = \left(\frac{k_1}{D_{AB}} \right)^{1/2}$$

↓
 solution

$$C_A(x) = C_1 e^{my} + C_2 e^{-my}$$

Boundary cond.

$$① C_A(0) = C_0$$

$$② \frac{dC_A}{dy} \Big|_{y=L} = 0$$

$$C_A(x) = C_0 \frac{\cosh m(L-y)}{\cosh mL}$$

40/100

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2. Steady state diffusion [1 dimensional]

$$x = -L, C_{A1} = C_L$$

$$x = L, C_{A2} = C_R$$

medium 2. reaction and A is consumed

$$\frac{dN_A}{dx} = 0$$

$$N_A = -D \frac{dC_A}{dx}, \quad \frac{d}{dx} \left(-D \frac{dC_A}{dx} \right) = 0 \rightarrow \text{so } \frac{d^2 C_A}{dx^2} = 0$$

$$\frac{dC_A}{dx} = C_1 \rightarrow C_A = C_1 x + C_2$$

Boundary conditions

$$\textcircled{1} C_L = C_1(-L) + C_2 \rightarrow C_L = -C_1 L + C_2$$

$$\textcircled{2} \phi_2 C_R = C_1(L) + C_2 \rightarrow \phi_2 C_R = C_1 L + C_2$$

$$C_1 = \frac{\phi_2 C_R - C_L}{2} \rightarrow 2-1$$

$$C_2 = \frac{\phi_2 C_R + C_L}{2} \rightarrow 1+2$$

$$C_A = \frac{\phi_2 C_R - C_L}{2} x + \frac{\phi_2 C_R + C_L}{2}$$

Flux

$$N_A = -D \frac{dC_A}{dx} = -D C_1 = -D \left(\frac{\phi_2 C_R - C_L}{2} \right)$$

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